Assessing Trait Interdependence in Penguins

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Abstract

We evaluate whether bill length can be predicted from other measured penguin physical characteristics in the Palmer Penguins dataset, discarding categorical variables due to their unbalanced distributions. Using linear regression, we compare a single-predictor model (flipper length) to a two-predictor model that adds body mass. Flipper length is a strong, statistically significant predictor of bill length, and adding body mass does not meaningfully improve the pre-existing linear model. We report coefficient estimates, confidence and prediction intervals, and a nested-model ANOVA to further analyze the addition of body mass in the model.

Introduction

Bill length plays a crucial role in how penguins catch and consume their prey. Understanding these measurements helps in studying their ecology and behavior. However, exact bill measurements may not always be available during surveys or exams, so it is useful to know whether other traits can reliably estimate bill length.

The Palmer Penguins dataset contains straightforward, meaningful measurements from three Antarctic penguin species. Its combination of continuous physiological variables (bill length, bill depth, flipper length, body mass) and simple categorical labels (species, island, sex, year) makes it ideal for exploratory data analysis and emphasizing model selection, particularly when accounting for multicollinearity among predictors. In this study, we examine whether bill length can be predicted from other physiological data, comparing a single-predictor model (flipper length) with a two-predictor model that includes body mass.

Literature Review

The Palmer Penguins dataset was originally collected by Dr. Kristen Gorman and the Palmer Station Long Term Ecological Research (LTER) program. The dataset includes biometric measurements from three species of penguins, Adélie, Chinstrap, and Gentoo, located in the Palmer Archipelago of Antarctica. The measurements, which include bill length, bill depth, flipper length, body mass, sex, and island, have supported a variety of analytical applications in recent years.

Early literature on the dataset emphasizes its value as an accessible introduction to multivariable analysis. Horst, Hill, and Gorman (2020) introduced the dataset as a replacement for the set in teaching visualization and classification, arguing that its ecological context and interpretable features make it more engaging for students. Their release of the dataset through the palmerpenguins R package led to several instructional papers and classroom demonstrations illustrating its use in exploratory data analysis (EDA), including pairwise scatter plots, distributional comparisons, and species separability based on morphology.

Later analyses using the dataset have explored machine learning classification methods. Researchers and instructors have demonstrated that species can be accurately predicted using algorithms such as logistic regression, k-nearest neighbors, decision trees, and random forests. These studies consistently find that flipper length, bill length, and body mass are among the most informative predictors, while categorical variables such as island and sex provide additional information for model performance. Several applied statistics courses and workshop papers show that the dataset lends itself well to model comparison, cross-validation workflows, and feature importance interpretation in tree-based models.

Another strand of literature focuses on the dataset as a clean example for teaching data preprocessing. Because the dataset contains missing values in several attributes, it provides a natural entry point for discussing imputation techniques, such as median substitution for continuous variables and mode substitution for categorical variables. Classroom exercises and published teaching notes frequently use the dataset to illustrate practice in handling missing data, assessing outliers, encoding categorical variables, and applying scaling methods.

Methodology

We used the R programming language to conduct our analysis, and used Google Colab for collaboration.

Upon extracting the data and observing the dimension, there are 344 rows and 9 columns, where one column is an ID column. Each row represents the information of a single penguin. Its species, habitat (island), bill length and depth (mm), flipper length (mm), body mass (g), sex, and year are recorded. We were first interested in seeing any type of correlation between the quantitative variables. To view this correlation, we computed a correlation matrix of the 5 variables: bill length, bill depth, flipper length, body mass, and year.

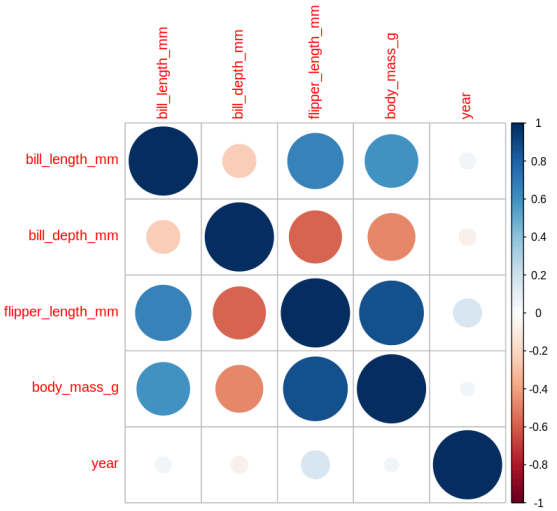


Figure 1: Correlation matrix of the quantitative variables.

As seen in Figure 1, all quantitative variables, except for year, have a moderate to strong correlation with each other, suggesting the presence of multicollinearity. Bill length and bill depth exhibit the weakest correlation among the physiological variables. Their scatterplot (Figure 2) shows two distinct clusters within the spread of points, indicating that a simple linear model would likely fail to capture the underlying relationship. The clusters could be formed due to the presence of species, as there are 3 different species in our dataset. Because bill depth lacks a clear linear relationship with bill length and appears structured by subgroups, we excluded bill depth from our analysis.

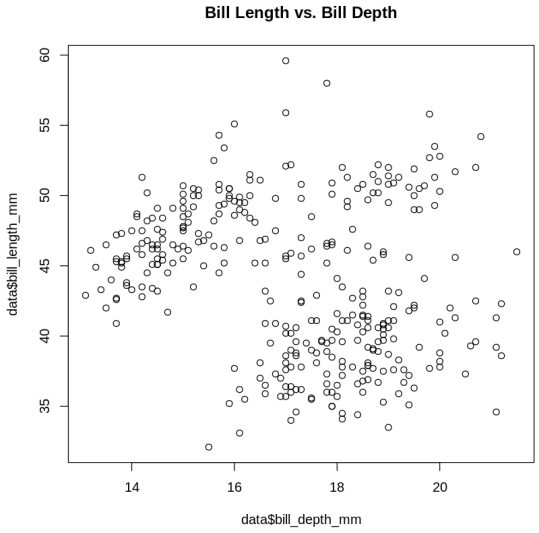
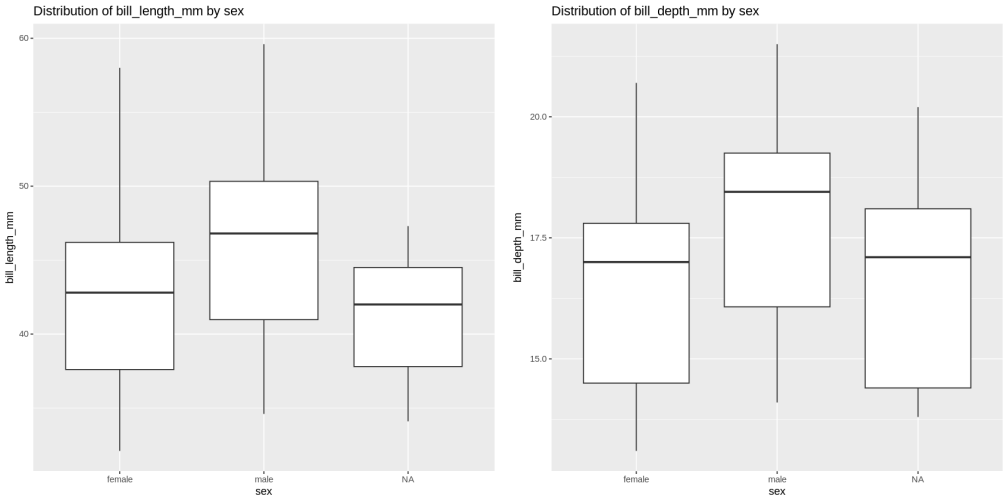


Figure 2: Scatter plot of bill length vs. bill depth.

We then inspected the categorical variables by counting the unique values for each category and found the distributions to be highly imbalanced. For example, species counts were Adelie = 152, Chinstrap = 68, Gentoo = 124; island counts were Biscoe = 168, Dream = 124, Torgersen = 52; and sex counts were female = 165, male = 168. While sex is roughly balanced, species and islands show noticeable imbalance (particularly Chinstrap and Torgersen), which could produce unstable subgroup estimates or biased model effects. For this reason, we did not choose the species or island variables as predictors. Although sex is roughly balanced and we briefly examined physiological summaries by sex (Figure 3), we did not pursue analyses using it because it would require additional preprocessing through two way factor analysis or ordinal encoding. However, it can be considered for future work.



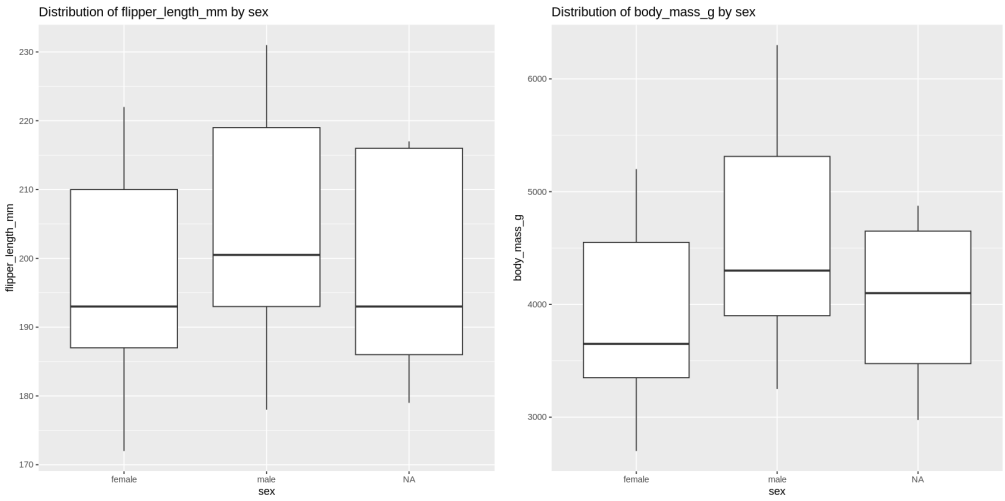


Figure 3: Box plots of physiological variables separated by sex.

There are 11 rows that are missing data, where all of them are not labeled for sex, and 2 of them (2/11) are missing data for physical characteristics. Since the missing rows make up a small percentage of roughly only 3% of our whole dataset (11/344), we decided to remove such observations from the analysis entirely. This leaves us with 333 remaining rows of data. After discarding the missing data, we proceed to the analysis.

Analysis

The analysis focuses on three variables of the Palmer Penguin dataset: bill length, flipper length, and body mass. Below, we present summary statistics for each variable to describe central tendency and spread before modeling:

Variable	Min.	Q1	Median	Mean	Q3	Max.
Bill Length (mm)	32.10	39.23	44.45	43.92	48.50	59.60
Flipper Length (mm)	172	190	197	200.9	213	231
Body Mass (g)	2700	3550	4050	4202	4750	6300

Figure 4: Summary statistics of bill length, flipper length, and body mass.

In this analysis, we address two research questions of interest: the first question is whether flipper length can be used to predict bill length in penguins, and the second question investigates whether the addition of the body mass variable, along with flipper length, will significantly improve our ability to predict bill length in penguins. The corresponding hypotheses are H_0: $\beta_{\text{flipper length}}=0$ versus H_1: $\beta_{\text{flipper length}}\neq 0$ for our first question of interest and H_0: $\beta_{\text{body mass}}=0$ versus H_1: $\beta_{\text{body mass}}\neq 0$ for the second question. A QQ plot was generated for the three variables of focus and for the fitted models of interest. The plots showed that assumptions of linearity are met, so we proceeded with our analysis.

To investigate the predictive ability of flipper length on bill length, the variables were fitted in a linear regression model with flipper length as the independent variable and bill length as the dependent

variable. A summary of the fitted model revealed a t-value of 15.691 for a t-test performed for the regression coefficient and p-value of 2 x 10^{-16} (Figure 5).

```
Call:
lm(formula = bill_length_mm ~ flipper_length_mm, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-8.6367 -2.6981 -0.5788  2.0663 19.0953

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     -7.21856    3.27175  -2.206    0.028 *
flipper_length_mm  0.25482    0.01624  15.691   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 4.148 on 331 degrees of freedom
Multiple R-squared:  0.4265,  Adjusted R-squared:  0.4248
F-statistic: 246.2 on 1 and 331 DF,  p-value: < 2.2e-16
```

Figure 5: Summary of the linear model (flipper length vs. bill length).

A large t-statistic value and a small p-value less than 0.05 led to our decision to reject the null hypothesis that $\beta_{\text{flipper length}}=0$. Our results indicate that $\beta_{\text{flipper length}}$ is statistically significant at an alpha level of 0.05, and the probability of the results being due to random chance is less than 5%. Therefore, flipper length can be used to predict bill length in penguins. As a follow-up, the cor function in R was used to calculate the correlation coefficient between flipper length and bill length, which was found to be 0.65. The correlation coefficient of 0.65 is closer to one, suggesting a moderate to strong positive linear relationship between the variables, and further supports the predictive ability of flipper length on bill length. The 95% confidence interval for the coefficients was calculated in R, and the coefficients provided in the summary were within the interval of (-13.6546, -0.7825) for the intercept and (0.2229, 0.2868) for the slope. So the equation of the linear model is bill length= -7.2186+0.2548 * flipper_length, where for every one mm increase in flipper length, the bill length is predicted to increase by 0.2548 mm.

Next, we assessed whether the addition of a variable, specifically body mass, would improve the predictive ability of flipper length to estimate bill length in penguins. As previously noted, we excluded the other variables, bill depth, sex, and island for our analysis because of data limitations, potential biases, or relationships that were highly non-linear. Therefore, body mass was chosen for our assessment. First, we found summary statistics regarding body mass alone on flipper length. The t-value for body mass alone is 13.28, and the p-value is 2 x 10^{-16} (Figure 6).

```
Call:
lm(formula = bill_length_mm ~ body_mass_g, data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-10.1652  -3.0664  -0.7672   2.2356  16.0371

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 2.715e+01  1.292e+00   21.02   <2e-16 ***
body_mass_g 4.003e-03  3.016e-04   13.28   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 4.424 on 331 degrees of freedom
Multiple R-squared:  0.3475,Adjusted R-squared:  0.3455
F-statistic: 176.2 on 1 and 331 DF,  p-value: < 2.2e-16
```

Figure 6: Summary of linear model (body mass vs. bill length).

The large t-statistic and small p-value that is less than 0.05 show that $\beta_{body\ mass}$ is statistically significant alone at an alpha level of 0.05. The body mass can be used alone to predict bill length in penguins. We then performed linear regression by fitting flipper length plus body mass as the independent variable and bill length as the dependent variable in a linear model. The summary of the fitted model showed a t-value of 6.822 for flipper length and 0.951 for body mass for the t-test (Figure 7). In addition, flipper length had a p-value of 4.31×10^{-11}, and for body mass, the p-value is 0.342 (Figure 7). Flipper length has a large t-statistic value and a small p-value that is less than 0.05, and body mass has a small t-statistic and a large p-value that is greater than 0.05, so we fail to reject the null hypothesis that $\beta_{body\ mass}=0$, which led to our decision that body mass does not add a significant predictive effect after accounting for flipper length.

```
Call:
lm(formula = bill_length_mm ~ flipper_length_mm + body_mass_g,
    data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-8.8251 -2.6432 -0.7281  2.0229 18.8227

Coefficients:
                   Estimate Std. Error t value Pr(>|t|)
(Intercept)       -3.9812079  4.7215540  -0.843    0.400
flipper_length_mm  0.2271747  0.0333014   6.822 4.31e-11 ***
body_mass_g        0.0005513  0.0005797   0.951    0.342
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 4.148 on 330 degrees of freedom
Multiple R-squared:  0.4281, Adjusted R-squared:  0.4246
F-statistic: 123.5 on 2 and 330 DF,  p-value: < 2.2e-16
```

Figure 7: Summary of Multilinear model (flipper length and body mass vs. bill length).

Additionally, the equation for the second research question is bill length= -3.9812 + 0.22717 * flipper_length + 0.00055 * body_mass. For every one mm increase in flipper length, the bill length should increase by 0.22717 mm, and for every g increase in body mass, the bill length increases by 0.00055. The confidence interval for this model is (41.472, 42.996), which means that we are 95% confident that the mean value for bill length with these characteristics is between 41.472 and 42.996mm. The prediction interval is (34.039, 50.430), which means that if we have a single penguin's bill length with these characteristics, it is likely between 34.039 and 50.430mm, with 95% probability.

A nested model analysis of variance (ANOVA) was performed where the larger nested model is $\text{bill_length_mm}=\beta_0+X_1\beta_{\text{flipper_length_mm}}+X_2\beta_{\text{body_mass_g}}+\epsilon$, containing both the flipper length and body mass predictors, and the smaller nested model is $\text{bill_length_mm}=\beta_0+X_1\beta_{\text{flipper_length_mm}}+\epsilon$. The p-value from the partial F-test is 0.3422, which is greater than the alpha level of 0.05 (Figure 8). Therefore, we fail to reject the null hypothesis that the models are equivalent, and this further supports our decision that the additional predictor, body mass, does not significantly improve our ability to predict bill length beyond flipper length. Considering the fact that the order of predictors may matter in statistical models, we found the nested model ANOVA for the linear model of adding body mass first, followed by flipper length, fitted on the dependent variable, bill length. The p-value from the partial F-test is 4.31×10^{-11} and demonstrates that flipper length as an additional predictor remains significant for predicting bill length even after accounting for body mass (Figure 9). However, for our research question, we’re focused on the predictive ability of body mass on bill length after accounting for flipper length.

A anova: 2 × 6

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	331	5693.889	NA	NA	NA	NA
2	330	5678.323	1	15.56557	0.9046045	0.3422461

Figure 8: Nested ANOVA summary with body mass as an additional predictor.

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	331	6479.080	NA	NA	NA	NA
2	330	5678.323	1	800.7561	46.53654	4.306811e-11

Figure 9: Nested ANOVA summary with flipper length as an additional predictor.

Conclusion

The analysis shows a clear relationship between flipper length and bill length, where flipper length is a strong predictor of bill length. For every one mm increase in flipper length, the bill length is predicted to increase by 0.2548 mm. When body mass is included in the model alongside flipper length, its small coefficient suggests that its addition to the linear regression model does not improve our ability to predict bill length. The nested-model ANOVA comparing the two-predictor model to the flipper-only model yields p=0.34, so adding body mass does not produce a significant reduction in variance.

Future Work

Future research could extend the present analysis in several meaningful ways. First, while our first study examined the predictive relationship between flipper length and bill length, additional variables, such as species, sex, and island location, could be incorporated to assess whether a multivariable model provides significantly greater explanatory power. Including ecological or demographic predictors may help account for variation between species and sex, both of which are known to influence characteristics. A very interesting variable to test would be island location and how it affects the characteristics of penguins from both the same species and different species. As we know, the Palmer penguins dataset includes data from penguins that live on different islands in the Palmer Archipelago of Antarctica. It would be interesting to see how island location can affect certain variables like body mass, flipper length, bill length, etc. It would also be interesting to see how adding a variable like species would affect the variable bias. The addition of such variables may make body mass significant, depending on collinearity. However, because our categorical predictors are imbalanced in this sample, including them could introduce bias or misleading inferences. Future work should either use a more balanced sample or apply techniques that address the varying weights among the unique species and island values for analysis.

Second, although we tested body mass as an additional predictor, the analysis could also benefit from a more systematic comparison of model selection techniques. For example, a stepwise regression approach would allow for stronger conclusions about variable importance. Model diagnostics could also be expanded to include residual normality tests, influence measures, and cross-validation to assess applicability.

Third, future work could explore non-linear modeling approaches. Polynomial regression, generalized additive model, or machine learning methods may capture patterns that linear regression cannot. These models could be evaluated using predictive performance metrics like adjusted R^2 to determine whether they offer substantial improvement of the linear regression model.

Finally, the presence of missing data in the dataset suggests an opportunity for further methodological refinement. Rather than listwise deletion or simple imputation, advanced imputation strategies, such as model-based imputation, could be applied to determine whether they influence model estimates or significance findings. Investigating these extensions would not only strengthen the reliability of the conclusions but also provide a deeper understanding of the predictors of penguin characteristics.

References

1. Chávez-Hoffmeister, Martín. (2020). Bill disparity and feeding strategies among fossil and modern penguins (GeoScienceWorld). pubs.geoscienceworld.org/paleosoc/paleobiol/article/46/2/176/587250/Bill-disparity-and-feeding-strategies-among-fossil.
2. Gorman, K. B., Williams, T. D., & Fraser, W. R. (2014). Ecological sexual dimorphism and environmental variability within a community of Antarctic penguins (Genus Pygoscelis). PLoS ONE, 9(3), e90081. https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0090081
3. Horst, A. M., Hill, A. P., & Gorman, K. B. (2020). palmerpenguins: Palmer Archipelago (Antarctica) penguin data (R package version 0.1.0) [Data set]. https://allisonhorst.github.io/palmerpenguins/articles/intro.html

Appendices

```
rows_with_na = data[rowSums(is.na(data)) > 0, ]
rows_with_na
```

A data.frame: 11 × 9

	X	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex	year
	<int>	<chr>	<chr>	<dbl>	<dbl>	<int>	<int>	<chr>	<int>
4	4	Adelie	Torgersen	NA	NA	NA	NA	NA	2007
9	9	Adelie	Torgersen	34.1	18.1	193	3475	NA	2007
10	10	Adelie	Torgersen	42.0	20.2	190	4250	NA	2007
11	11	Adelie	Torgersen	37.8	17.1	186	3300	NA	2007
12	12	Adelie	Torgersen	37.8	17.3	180	3700	NA	2007
48	48	Adelie	Dream	37.5	18.9	179	2975	NA	2007
179	179	Gentoo	Biscoe	44.5	14.3	216	4100	NA	2007
219	219	Gentoo	Biscoe	46.2	14.4	214	4650	NA	2008
257	257	Gentoo	Biscoe	47.3	13.8	216	4725	NA	2009
269	269	Gentoo	Biscoe	44.5	15.7	217	4875	NA	2009
272	272	Gentoo	Biscoe	NA	NA	NA	NA	NA	2009

```
library(tidyr)
data=data%>%drop_na(bill_length_mm,sex)
```

Double checking for NA values

```
rows_with_na = data[rowSums(is.na(data)) > 0, ]
rows_with_na
```

A data.frame: 0 × 9

X	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex	year
<int>	<chr>	<chr>	<dbl>	<dbl>	<int>	<int>	<chr>	<int>

```r
penguins <- read.csv("penguins.csv")


penguins <- penguins[, -1]

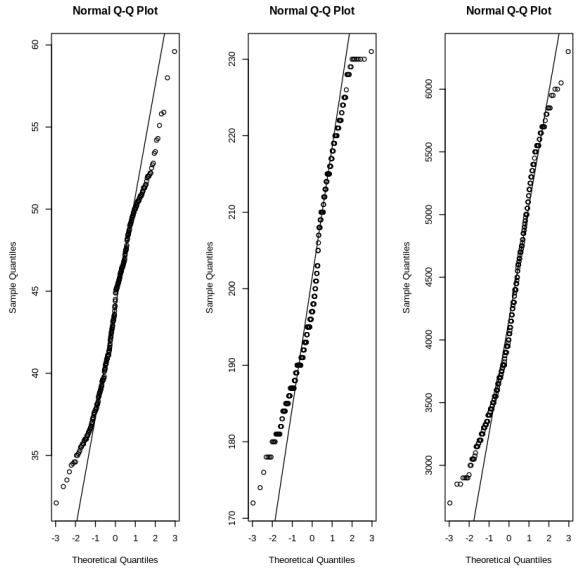

penguins_clean <- penguins %>%
  drop_na(bill_length_mm, flipper_length_mm, body_mass_g, sex)

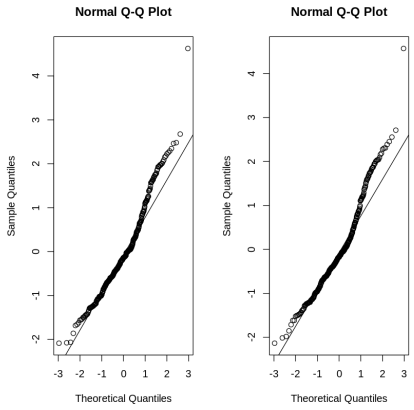
# QQ plots of the 3 variables
par(mfrow = c(1, 3))
qqnorm(penguins_clean$bill_length_mm); qqline(penguins_clean$bill_length_mm)
qqnorm(penguins_clean$flipper_length_mm); qqline(penguins_clean$flipper_length_mm)
qqnorm(penguins_clean$body_mass_g); qqline(penguins_clean$body_mass_g)
par(mfrow = c(1, 1))

# Regression models
mod1 <- lm(bill_length_mm ~ flipper_length_mm, data = penguins_clean)
mod2 <- lm(bill_length_mm ~ flipper_length_mm + body_mass_g, data = penguins_clean

# Residual QQ Plots
res1 <- rstandard(mod1)
res2 <- rstandard(mod2)

par(mfrow = c(1, 2))
qqnorm(res1); qqline(res1)
qqnorm(res2); qqline(res2)
par(mfrow = c(1, 1))
```






```r
fit12=lm(bill_length_mm~flipper_length_mm+body_mass_g, data=data)
summary(fit12)
```

Show hidden output

uation of linear model: bill_length_mm= -3.9812+0.22717 * flipper_length_mm + 0.00055 * body_mass_g

```r
fit12$coefficients
```

Show hidden output

```r
summary(fit12)$coefficients["body_mass_g", 3]   # t value
summary(fit12)$coefficients["body_mass_g", 4]   # p value
new <- data.frame(flipper_length_mm = 193, body_mass_g = 4300)
predict(fit12, new, interval = "confidence", level = 0.95) #confidence interval
predict(fit12, new, interval = "prediction", level = 0.95) #prediction interval
```

```r
fit21=lm(bill_length_mm~body_mass_g+flipper_length_mm, data=data)
```

```r
anova(fit2, fit21)
```

A anova: 2 × 6

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	331	6479.080	NA	NA	NA	NA
2	330	5678.323	1	800.7561	46.53654	4.306811e-11